

Upheld platform calculations

Product: Upheld Clinical Revenue Integrity

Audience: Agency QA, revenue cycle, pilot evaluation, engineering

Version: chart-scan-v3 + PDGM payment intelligence (Phase 2)

Status: Advisory — **not** a certified CMS HIPPS grouper or remittance engine

This document explains **every number** the platform shows: readiness scores, expected period payment, capture (recover), protect (at risk), and LUPA-related estimates.

1. What the platform is calculating

Output	Question it answers
Submission readiness (0–100)	How “clean” is this packet for submission?
Expected period total (\$)	What should full PDGM period pay be for this chart (advisory)?
Capture / recoverable (\$)	What upside if documentation/coding is fixed?
Protect / at risk (\$)	What is exposed if submitted as-is?
Path to 80+	Which open findings drag readiness, and in what order to fix?
Batch would-have-caught	On already-processed claims, would we have flagged the denial/LUPA?

All dollar figures are **advisory**, anchored to **CMS CY 2026 Home Health PPS** national rates, then adjusted by a **chart-specific case-mix model**. They are **not** final claim amounts.

2. CMS rate anchors (source of truth for \$)

2.1 National standardized 30-day period payment

Parameter	Value	Notes
CMS payment year	2026	CY 2026 HH PPS Final Rule (CMS-1828-F) family
National 30-day base (quality data submitters)	\$2,038.22	Used as nationalBase
National 30-day base (no quality data)	\$1,998.41	Available in code; not default for estimates

Important: Under PDGM, agencies are paid in **30-day periods**. A physician certification period is often **60 days** and may include **two** 30-day payments. Upheld’s “expected period total” is for **one 30-day PDGM payment period**, not automatically $2 \times$ that amount.

2.2 Per-visit rates (LUPA path)

Used when modeling low-utilization (LUPA) payment instead of full period:

Discipline	CY 2026 national per-visit (approx.)
Skilled nursing (SN)	\$176.96
Physical therapy	\$193.42
Occupational therapy	\$194.74
Speech-language pathology	\$210.25
Medical social services	\$283.64
Home health aide	\$80.12

2.3 Illustrative LUPA gap (national, mid-band)

$$LUPA\ gap_{national} \approx \$2,038.22 - (5 \times \$176.96) \approx \textbf{\$1,153}$$

Threshold “5” is a **mid-band family assumption**, not every HIPPS-specific LUPA threshold.

3. Expected period total (chart-specific)

3.1 Formula

$$\text{Expected period payment} = \text{National base} \times \text{Case-mix weight} \times \text{Wage index}$$

Term	Default	Source
National base	\$2,038.22	CMS CY 2026 national standardized 30-day rate
Case-mix weight	Chart-inferred	Packet signals and/or sparse HIPPS table
Wage index	1.0 (national)	Override with env AGENCY_WAGE_INDEX

Example: $\$2,038.22 \times 1.18 \times 1.00 \approx \textbf{\$2,405}$

3.2 How case-mix weight is built

Method A — HIPPS table (when known): If the chart contains a HIPPS code and that code is in Upheld’s small lookup table, that weight is used.

Method B — Signal model (usual path):

$$\text{Weight} = W_{family} \times F_{comorbidity} \times F_{functional} \times F_{timing} \times F_{admission}$$

Weights are **clamped** to roughly **0.55 – 2.8** so outliers do not explode.

Clinical group family base weights

Inferred from narrative / diagnoses (wound, neuro, MMTA, complex, rehab, behavioral, default):

Family	Relative base weight
COMPLEX	1.32
WOUND	1.18
NEURO	1.14
MS_REHAB	1.10
MMTA	1.02
DEFAULT	1.00
BEHAVIORAL	0.96

Comorbidity band

Band	Factor	How inferred (examples)
NONE	1.00	Explicit “no secondary” / blank M1023 language
LOW	1.06	Some secondary dx / 2+ chronic markers
HIGH	1.14	Many markers (CHF, COPD, CKD, stage 3–4 ulcer, etc.) or “high comorbidity”
UNKNOWN	1.02	Insufficient signals

Functional band

Band	Factor	How inferred
LOW	0.94	Independent / low OASIS-style scores
MEDIUM	1.00	Mid scores or unclear
HIGH	1.10	High M1800–M1860 / GG impairment or total assist language
UNKNOWN	1.00	Default

Timing

Band	Factor	How inferred
EARLY	1.03	“Early”, SOC / first period language
LATE	0.97	“Late”, recert / second period
UNKNOWN	1.00	Default

Admission source

Band	Factor	How inferred
COMMUNITY	1.00	Community admission / no recent inpatient
INSTITUTIONAL	1.05	Hospital / SNF / institutional language
UNKNOWN	1.00	Default

3.3 Confidence

Confidence	Meaning
HIGH	HIPPS hit in table, or many packet signals known
MEDIUM	Partial signals
LOW	Sparse chart → weight near national

3.4 What expected period total is not

- Not final remittance
- Not full CMS HIPPS weight file (thousands of cells)
- Not CBSA wage index unless you set AGENCY_WAGE_INDEX
- Not automatically 60-day cert total (would be ~two periods if both paid)

The report **basis line** lists the exact factors used for that scan.

4. Finding dollars (per issue)

Each finding may carry an **estimatedImpact** (\$) and an **impactType**:

impactType	Meaning on report
EXPOSURE	Protect — at risk if submitted as-is
RECOVERY	Capture — recoverable if fixed
NEUTRAL	Integrity/process issue without \$

4.1 National defaults (before chart scaling)

Defaults are fractions of the **national** \$2,038.22 base (rounded):

Concept	Approx. national \$	Rule of thumb
F2F denial share	~45% of period	Denial-class
Homebound	~38% of period	Denial-class
Certification gap	~22% of period	Denial-class
Order / signature	~18% of period	Denial-class
Case-mix undercoding	~11% of period	Capture
Comorbidity miss	~5% of period	Capture
LUPA full-period gap	~\$1,153	National mid-band
Late NOA	~period/30 × 2 days	Exposure
Wound / skilled need	~3–6% of period	Mixed

4.2 Scaling to this chart's period

After the chart's **expected period payment** E is known:

$$Finding \$_{chart} = \min(E, Finding \$_{national} \times E / \$2,038.22)$$

So richer case-mix charts get **proportionally larger** advisory \$; thin charts stay closer to national.

4.3 LUPA finding special case

When LUPA risk is **likely**, the LUPA finding impact is set toward:

$$LUPA\ gap_{chart} \approx E - (5 \times SN\ per\ visit \times wage\ index)$$

Visit counts still come from the chart (e.g. “3 skilled visits”). Threshold remains a **family mid-band** unless expanded with full HIPPS LUPA tables.

5. Capture (recover) and protect (at risk) totals

5.1 Protect — revenue at risk

Only findings with **EXPOSURE** and positive \$:

- **Denial-ceiling class** (F2F, homebound, certification, physician order/signature, and listed categories): take the **maximum** once — you cannot stack full denials of the same claim.
- **Other exposure** (LUPA gap, NOA, some clinical docs): **sum**.
- Final protect total:

$$Protect = \min(\max(denial\ ceiling, other\ exposure\ sum), E)$$

where E = expected period payment for that chart.

5.2 Capture — recoverable if fixed

Only findings with **RECOVERY**:

$$Capture = \min(\sum RECOVERY \$, E)$$

Capture and protect are **separate paths**. Do **not** add them to invent a new “total claim.” Think of them as:

- **Protect**: how much of E is in jeopardy as-is

- **Capture:** how much additional/supported payment might be unlocked if coding/docs improve

5.3 Report layout (three money cards)

Card	Value
Expected period total	E (chart-specific)
Recoverable if fixed	Capture
At risk if submitted	Protect

6. Submission readiness score (0–100)

6.1 Severity penalties

Severity	Penalty points
CRITICAL	22
HIGH	12
MEDIUM	6
LOW	2

6.2 Module scores

For each module $m \in \{\text{CLINICAL}, \text{COMPLIANCE}, \text{REVENUE}\}$:

$$\text{Module score}_m = 92 \text{ if no findings in } m; \text{ else } \max(0, \min(100, 100 - \sum \text{penalties in } m))$$

6.3 Overall readiness

$$\text{Readiness} = \text{round}(0.35 \times \text{Clinical} + 0.40 \times \text{Compliance} + 0.25 \times \text{Revenue})$$

Weights emphasize **compliance** (orders, F2F, homebound, certification) slightly more than clinical or revenue modules.

6.4 Live readiness vs analysis snapshot

Concept	Definition
Analysis snapshot	Score at first complete analysis (stored as originalReadiness)
Live readiness	Recomputed from OPEN findings only

When a finding is **Resolved** or **Dismissed**, it **no longer penalizes** live readiness. The ring and the path-to-80 panel both use **live** readiness so they stay aligned.

6.5 Path to readiness 80+

Gate: **READINESS_GATE = 80**. If live readiness < 80:

- Rank OPEN findings by severity penalty (compliance slightly boosted).
- Show ordered correction steps (suggested fix + CMS ref when available).
- Project readiness after each successive fix.
- Highlight the step that **clears 80**.

7. Multi-pass analysis (how findings are produced)

Pass	Module	Examples
1	Clinical Integrity	OASIS consistency, wound progression, skilled need narrative, reassessment
2	Compliance Intelligence	F2F, homebound, certification, signatures, CoP-related gaps
3	Revenue Intelligence	LUPA volume, comorbidity/coding support, NOA timing, PDGM undercoding

Optional: LUPA structured pass rewrites/replaces LUPA findings from visit counts. Optional: LLM enrichment (off by default) may add findings; dollars still capped by E. Deterministic rules use CMS references (e.g. 42 CFR §424.22 for F2F) for citations — not for remittance math.

8. Retrospective batch (“would-have-caught”)

For already-processed claims with labeled outcomes (DENIED, LUPA, TAKEBACK, PAID_CLEAN, etc.):

- Run the same multi-pass analysis per claim.
- Compare findings to known outcome + known reason.
- **Caught** = material HIGH/CRITICAL (or \$ exposure) finding that matches the outcome family / reason keywords.
- **Catch rate** = caught ÷ adverse labeled claims.
- **Loss on caught** = sum of labeled knownLossUsd on claims marked caught.

This measures integrity detection quality against history — still not remittance simulation.

9. Worked example (illustrative)

Chart signals: Wound-heavy, early timing, community admission, high comorbidity language, low visit volume (LUPA risk).

- Weight $\approx 1.18 \times 1.14 \times 1.00 \times 1.03 \times 1.00 \approx \mathbf{1.39}$ (example)
- Expected period $\approx 2038.22 \times 1.39 \times 1.0 \approx \mathbf{\$2,830}$
- F2F finding national ~\$917 → scaled ~\$1,275 (still ≤ E)
- Denial-ceiling findings take **max**, not sum
- LUPA gap $\approx E - 5 \times 176.96$
- Protect = min(max(denial ceiling, other exposure), E)
- Capture = sum of recovery findings, capped at E
- Readiness from open severity penalties by module

Actual numbers on a live report depend on the exact packet text.

10. Configuration knobs

Env var	Effect
AGENCY_WAGE_INDEX	Multiplier on expected period (default 1.0)
FREE_SCAN_* rate limits	Abuse controls only — not dollars
XAI_API_KEY / LLM	Optional finding enrichment — still capped by E

11. Explicit non-claims (compliance language)

Upheld **does not**:

- Issue a certified HIPPS code as a grouper of record
- Replace the Medicare claims system or MAC remittance
- Guarantee payment, denial, or LUPA outcome
- Train foundation models on customer PHI without separate consent

Upheld **does**:

- Surface documentation, compliance, and revenue-integrity **risks**

- Anchor \$ to **CMS national rates** and a **transparent case-mix model**
- Require **human review** before claim action

12. Roadmap toward remittance-grade uniqueness

Stage	Capability
Today (Phase 2)	Signal-based case-mix × national base × optional wage index; scaled capture/protect
Next	Full CMS HIPPS weight file + HIPPS-specific LUPA thresholds
Later	CBSA wage index by agency address; agency-specific fee schedule imports

13. Where to see calculations in product

Surface	What shows
Chart scan report	Expected period total, capture, protect, basis line, readiness, path to 80
JSON /api/scans/[token]	expectedPeriodPayment, paymentEstimate, scores, findings
CSV export	Meta rows including readiness and \$ fields
Trust page / CMS rate card	National anchors and per-visit reference
Retrospective batch	Would-have-caught rates vs labeled claim history

14. Summary one-liner

Readiness = weighted integrity of the chart.

Expected period \$ = CMS national 30-day base × chart-inferred case-mix × wage index.

Protect = exposure \$ (denials max'd once, other exposure summed), capped at expected period.

Capture = recovery \$ summed, capped at expected period.

None of this is a remittance — it is Clinical Revenue Integrity decision support.

Document generated for Upheld / Humble Haus Ventures. Implementation: src/domain/chart-scan/ (knowledge.ts, scoring.ts, pdgm-payment.ts, readiness-path.ts, pipeline.ts).